

# AA4 – e-NFAs

## (soluções)

1

A spontaneous transition from a state A to a state B in a  $\epsilon$ -NFA means \_\_\_\_\_ and \_\_\_\_\_.   
(0/1 Ponto)

*For the first missing snippet select one of the first two options and for the second snippet, from the remaining options, select the correct one(s).*

- ☐ to process the current symbol of the input and ignore its value
- ☒ to not process the current symbol
- ☒ stay in A
- ☐ simply move to B
- ☒ build a new path moving to state B
- ☒ recursively build new paths depending on spontaneous transitions of B

2

The conversion of a  $\epsilon$ -NFA a DFA will result in a DFA with \_\_\_\_\_ states.  
(1/1 Ponto)

- ☐ *exactly  $2 \times n$*
  - ☐ *at most  $2 \times n$*
  - ☐ *exactly  $2^n$*
  - ☒ *at most  $2^n$*
  - ☐ *possibly more than  $2^n$*
-

3

We can use a DFA to represent a problem requesting a  $\epsilon$ -NFA and, similarly, we can use a  $\epsilon$ -NFA to represent when a NFA is requested.

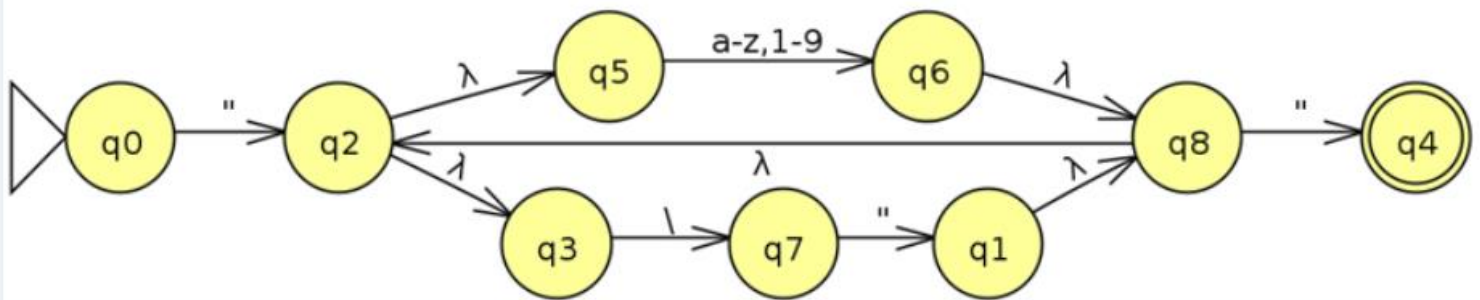
(0/1 Ponto)

- ☐ The sentence is true.
- ☐ The sentence is false.
- ☒ The sentence is true for the first part and false for the second part..
- ☐ The sentence is false for the first part and true for the second part.

4

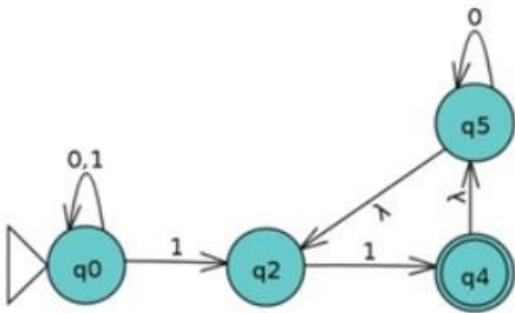
Which of the following strings are accepted by the automaton?

(1/1 Ponto)

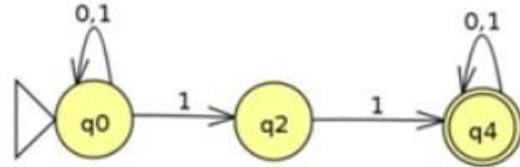


- ☒ "myfirsttest"
- ☐ "yet"another"test"
- ☒ "2more\"2go"
- ☐ "a\"2"
- ☐ ""
- ☐ <empty string>

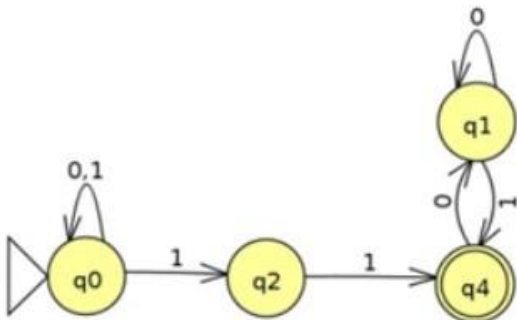
Which of the following automata represent the same language?  
(0/1 Ponto)



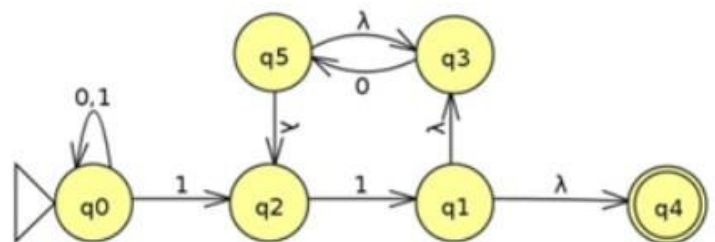
a)



b)



c)

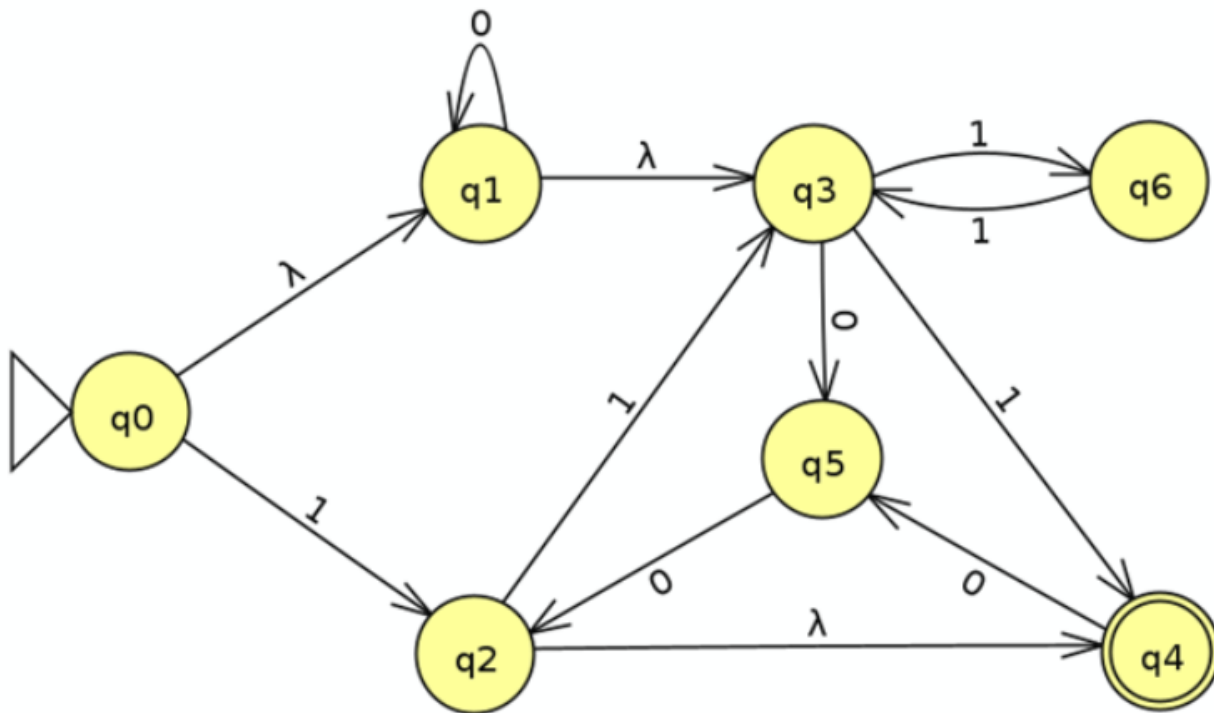


d)

Select the sentences that better describe the language represented by this  $\epsilon$ -NFA:

"Strings over the alphabet  $\{0,1\}$  with a prefix of an arbitrary number of zeros or '11', and the remainder is composed by... 

(0/1 Ponto)



- ☒ any number of 1's
- ☐ an odd number of 1's
- ☐ the 1's can be mixed with a maximum of two contiguous 0's
- ☒ the 1's can be mixed with an even number of 0's
- ☐ and always ends with a 1